

SPECIAL SUMMER INTEGRATED LEARNING PROGRAMME MATHEMATICS

Sequences and Series

Week 4 Day 2 PM



SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN

OUTLINE

- 1 Sequences and Summation Notations
- 2 Arithmetic Sequences
- 3 Geometric Sequences and series
- 4 The Binomial Theorem
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Roughly speaking, a sequence is an infinite ordered list of numbers. The numbers in the sequence are often written as $a_1, a_2, a_3, \dots$. The dots mean that the list continues forever.

DEFINITION:

A **sequence** is a function $a : \mathbb{N}^* \rightarrow \mathbb{R}$. The **terms** of the sequence are the function values

$$a(1), a(2), a(3), \dots, a(n), \dots$$

We usually write a_n instead of the function notation $a(n)$ and denote the sequence as

$$(a_n)_{n=1}^{\infty} \quad \text{or} \quad (a_1, a_2, a_3, \dots).$$

a_n is called the ***n*th term** of the sequence.

Note that by default a sequence is an infinite list. For finite lists of numbers, they are called *finite sequences*. Restricted

SEQUENCES - EXAMPLE

E.g.:

$$((-1)^n)_{n=0}^{\infty} = (1, -1, 1, -1, 1, \dots).$$

Remark: It's not a strict rule for sequences to begin with index $n = 1$. The sequence above begins at $n = 0$, which is also fine.

E.g.:

$$\left(\frac{n^3 + 3n + 5}{n^2 + 1} \right)_{n=1}^{\infty} = \left(\frac{9}{2}, \frac{19}{5}, \frac{41}{10}, \dots \right).$$

E.g.:

$$(n\text{-th prime number})_{n=1}^{\infty} = (2, 3, 5, 7, 11, 13, 17, \dots).$$

E.g.:

$$(n!)_{n=0}^{\infty} = (1, 1, 2, 6, 24, 120, 720, \dots),$$

where $n! = 1 \times 2 \times 3 \times \dots \times n$, with the convention $0! := 1$.

FINDING THE TERMS OF A SEQUENCE

If the formula of the general n th term of a sequence $(a_n)_{n=1}^{\infty}$ is given, say $a_n = f(n)$, then to find a_k , we simply evaluate $f(n)$ at $n = k$.

E.g.: Given $a_n = 2n - 1$. Then $(a_n)_{n=1}^{\infty} = (1, 3, 5, 7, 9, \dots)$ and $a_{100} = 2(100) - 1 = 199$.

E.g.: Given $b_n = n^2 - 1$. Then $(b_n)_{n=1}^{\infty} = (0, 3, 8, 15, 24, \dots)$ and $b_{100} = 100^2 - 1 = 9999$.

E.g.: Given $c_n = \frac{n}{n+1}$. Then $(c_n)_{n=1}^{\infty} = \left(\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots\right)$ and $c_{100} = \frac{100}{101}$.

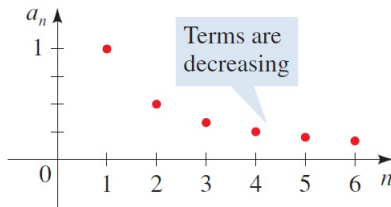
E.g.: Given $d_n = \frac{(-1)^n}{2^n}$. Then $(d_n)_{n=1}^{\infty} = \left(-\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}, \frac{1}{16}, -\frac{1}{32}, \dots\right)$ and $d_{100} = \frac{(-1)^{100}}{2^{100}} = \frac{1}{2^{100}}$.

Note that the presence of $(-1)^n$ in the sequence has the effect of making successive terms alternately negative and positive.

THE GRAPH OF A SEQUENCE

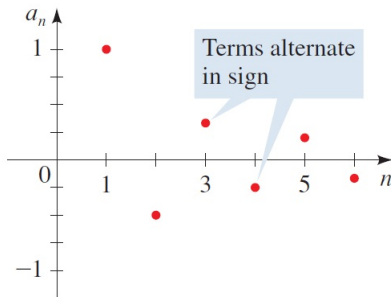
Since a sequence is a function with domain $\mathbb{N}$ and codomain $\mathbb{R}$, we can plot its graph just like any other functions with domain being a subset of $\mathbb{R}$ and codomain $\mathbb{R}$. Since the domain is $\mathbb{N}$, the graph consists of discrete points instead of a curve.

E.g.: Below is the graph of $\left(\frac{1}{n}\right)_{n=1}^{\infty}$.



THE GRAPH OF A SEQUENCE - EXAMPLE

E.g.: Below is the graph of $\left(\frac{(-1)^{n+1}}{n}\right)_{n=1}^{\infty}$.



You can use this [Geogebra applet](#) to graph any sequences by entering the formula of the n th term.

RECURSIVELY DEFINED SEQUENCES

Instead of the n th term being defined in terms of n , some sequences are naturally defined recursively. More precisely, the first k terms of the sequence are given explicitly and the n th term of a sequence is a function of the k preceding terms, i.e.

$$a_{n+k} = f(a_{n+k-1}, a_{n+k-2}, \dots, a_{n+1}, a_n).$$

for all $n \in \mathbb{N}^*$. A sequence defined in this way is called **recursive**.

E.g.: A sequence is defined recursively by $a_1 = 1, a_2 = -1$ and

$$a_{n+2} = 3a_{n+1}(a_n + 2).$$

Then

$$a_3 = 3a_2(a_1 + 2) = 3(-1)(1 + 2) = -9,$$

$$a_4 = 3a_3(a_2 + 2) = 3(-9)(-1 + 2) = -27,$$

$$a_5 = 3a_4(a_3 + 2) = 3(-27)(-9 + 2) = 567,$$

$$a_6 = 3a_5(a_4 + 2) = 3(567)(-27 + 2) = -42525,$$

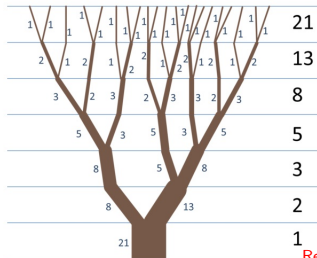
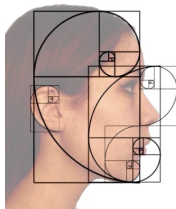
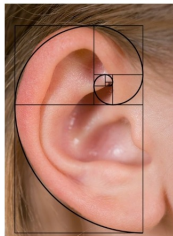
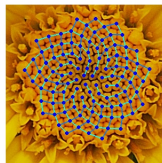
$\vdots$

ACTIVITY 1

Fibonacci sequence is a sequence that can be observed in nature and has many applications. It is defined recursively as follows:

$$F_1 = F_2 = 1, \quad F_{n+2} = F_{n+1} + F_n, \quad \text{for } n \geq 1.$$

- (a) Find the first seven terms of the sequence.
- (b) Use Excel to find F_{100} .
- (c) Plot $(F_n)_{n=0}^{\infty}$. What do you observe as $n \rightarrow \infty$?
- (d) Define the new sequence $r_n := \frac{F_n}{F_{n-1}}$ for $n \geq 2$. Use Excel to find the first 99 terms. What do you observe?
- (e) Change the values of F_1 and F_2 as you like (they can be negative or irrational) as long as both are not zero. What do you observe about the sequence $(r_n)_{n=0}^{\infty}$?



ACTIVITY 1: SOLUTION

(a) $F_1 = F_2 = 1$

$$F_3 = F_2 + F_1 = 1 + 1 = 2,$$
$$F_4 = F_3 + F_2 = 2 + 1 = 3,$$
$$F_5 = F_4 + F_3 = 3 + 2 = 5,$$
$$F_6 = F_5 + F_4 = 5 + 3 = 8,$$
$$F_7 = F_6 + F_5 = 8 + 5 = 13.$$

(b) We can set up the Excel sheet as follows:

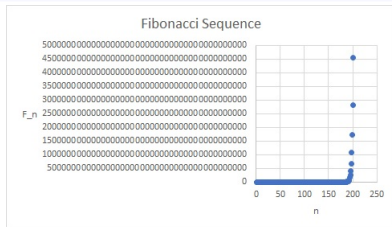
	A	B
1	n	F_n
2	1	1
3	2	1
4	3	=B2+B3
5	4	
6	5	

and easily compute that

$F_{100} \approx 354, 224, 848, 179, 262, 000, 000$ (up to 15 significant figures, which is the capacity of Excel). The actual value is 354, 224, 848, 179, 261, 915, 075.

ACTIVITY 1: SOLUTION

(c)



It seems like F_n goes to infinity as $n \rightarrow \infty$.

(d) It seems like r_n becomes the constant 1.618034 when n is large.

(e) The behavior of r_n when n is large stays unchanged.

Fun fact: By linear algebra, it can be shown that

$$F_n = A \left(\frac{1 + \sqrt{5}}{2} \right)^n + B \left(\frac{1 - \sqrt{5}}{2} \right)^n,$$

where A, B are constants that depend on F_0, F_1 . With this formula, we can easily prove all the observations in this activity.

THE PARTIAL SUM OF A SEQUENCE

Quite often, we are interested in adding the terms of a sequence. This leads to the following definition.

DEFINITION:

For the sequence $(a_1, a_2, a_3, \dots)$, the **partial sums** are

$$S_1 = a_1$$

$$S_2 = a_1 + a_2$$

$$S_3 = a_1 + a_2 + a_3$$

$$\vdots$$

$$S_n = a_1 + a_2 + a_3 + \cdots + a_n$$

$$\vdots$$

S_n is called the **n th partial sum**. The sequence $(S_1, S_2, S_3, \dots)$ is called the **sequence of partial sums corresponding to the sequence** $(a_n)_{n=1}^{\infty}$.

THE PARTIAL SUM OF A SEQUENCE - EXAMPLE

E.g.: Suppose $a_n = \frac{1}{2^n}$ for $n \in \mathbb{N}^*$. Then

$$\begin{aligned} S_1 &= \frac{1}{2} &= \frac{1}{2} \\ S_2 &= \frac{1}{2} + \frac{1}{4} &= \frac{3}{4} \\ S_3 &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} &= \frac{7}{8} \\ S_4 &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} &= \frac{15}{16} \\ &\vdots & \end{aligned}$$

We can show that in fact $S_n = \frac{2^n - 1}{2^n}$. This follows immediately from the partial sum formula of geometric sequences, which we'll see in a moment.

THE PARTIAL SUM OF A SEQUENCE - EXAMPLE

E.g.: Suppose $a_n = \frac{1}{n} - \frac{1}{n+1}$ for $n \in \mathbb{N}^*$. Then

$$S_1 = \left(1 - \frac{1}{2}\right) = 1 - \frac{1}{2}$$

$$S_2 = \left(1 - \cancel{\frac{1}{2}}\right) + \left(\cancel{\frac{1}{2}} - \frac{1}{3}\right) = 1 - \frac{1}{3}$$

$$S_3 = \left(1 - \cancel{\frac{1}{2}}\right) + \left(\cancel{\frac{1}{2}} - \cancel{\frac{1}{3}}\right) + \left(\cancel{\frac{1}{3}} - \frac{1}{4}\right) = 1 - \frac{1}{4}$$

$$S_4 = \left(1 - \cancel{\frac{1}{2}}\right) + \left(\cancel{\frac{1}{2}} - \cancel{\frac{1}{3}}\right) + \left(\cancel{\frac{1}{3}} - \cancel{\frac{1}{4}}\right) + \left(\cancel{\frac{1}{4}} - \frac{1}{5}\right) = 1 - \frac{1}{5}$$

⋮

Hence, we deduce that $S_n = 1 - \frac{1}{n+1}$.

ACTIVITY 2

(a) Find the first four partial sums of the following sequences:

(i) $(1, 3, 5, 7, \dots)$

(ii) $((-1)^n)_{n=1}^{\infty}$

(b) Find the formula of the n th partial sum of the sequence with the following formula of its n th term:

(i) $a_n = \sqrt{n} - \sqrt{n+1}$

(ii) $a_n = \ln\left(\frac{n}{n+1}\right)$

ACTIVITY 2: SOLUTION

(a) (i)

$$S_1 = 1,$$

$$S_2 = 1 + 3 = 4,$$

$$S_3 = 1 + 3 + 5 = 9,$$

$$S_4 = 1 + 3 + 5 + 7 = 16.$$

(ii)

$$S_1 = (-1)^1 = -1,$$

$$S_2 = (-1)^1 + (-1)^2 = 0,$$

$$S_3 = (-1)^1 + (-1)^2 + (-1)^3 = -1,$$

$$S_4 = (-1)^1 + (-1)^2 + (-1)^3 + (-1)^4 = 0.$$

ACTIVITY 2: SOLUTION

(b) (i)

$$S_1 = (\sqrt{1} - \sqrt{2}) \quad = 1 - \sqrt{2}$$

$$S_2 = (\sqrt{1} - \cancel{\sqrt{2}}) + (\cancel{\sqrt{2}} - \sqrt{3}) \quad = 1 - \sqrt{3}$$

$$S_3 = (1 - \cancel{\sqrt{2}}) + (\cancel{\sqrt{2}} - \cancel{\sqrt{3}}) + (\cancel{\sqrt{3}} - \sqrt{4}) \quad = 1 - \sqrt{4}$$

$$S_4 = (1 - \cancel{\sqrt{2}}) + (\cancel{\sqrt{2}} - \cancel{\sqrt{3}}) + (\cancel{\sqrt{3}} - \cancel{\sqrt{4}}) + (\cancel{\sqrt{4}} - \sqrt{5}) \quad = 1 - \sqrt{5}$$

$\vdots$

Hence, we deduce that $S_n = 1 - \sqrt{n+1}$.

ACTIVITY 2: SOLUTION

(b) (i) Note that $\ln\left(\frac{n}{n+1}\right) = \ln n - \ln(n+1)$, so

$$S_1 = (\ln 1 - \ln 2) = -\ln 2$$

$$S_2 = (\ln 1 - \cancel{\ln 2}) + (\cancel{\ln 2} - \ln 3) = -\ln 3$$

$$S_3 = (\ln 1 - \cancel{\ln 2}) + (\cancel{\ln 2} - \cancel{\ln 3}) + (\cancel{\ln 3} - \ln 4) = -\ln 4$$

$$S_4 = (\ln 1 - \cancel{\ln 2}) + (\cancel{\ln 2} - \cancel{\ln 3}) + (\cancel{\ln 3} - \cancel{\ln 4}) + (\cancel{\ln 4} - \ln 5) = -\ln 5$$

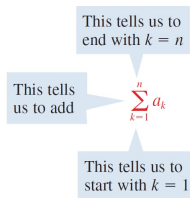
$\vdots$

Hence, we deduce that $S_n = -\ln(n+1)$.

THE SIGMA NOTATION

It's convenient to introduce the **summation notation**/**sigma notation** to denote sums of terms:

$$\sum_{k=1}^n a_k := a_1 + a_2 + a_3 + \cdots + a_n.$$



k is called the **index of summation**/**summation variable**, its increment is always 1. It's a dummy variable, you can replace it with other letters/symbols.

THE SIGMA NOTATION - EXAMPLE

To evaluate a sigma notation, simply substitute k by the starting value into a_k , then increase it by 1 and repeat the process until k becomes the ending value. After that, sum up all the values.

E.g.:

$$\sum_{k=1}^5 k^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55.$$

E.g.:

$$\sum_{j=3}^5 \frac{1}{j} = \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{47}{60}.$$

E.g.:

$$\sum_{k=-1}^3 k = (-1) + 0 + 1 + 2 + 3 = 5.$$

E.g.:

$$\sum_{i=1}^6 2 = 2 + 2 + 2 + 2 + 2 + 2 = 12.$$

THE SIGMA NOTATION - EXAMPLE

We often use the sigma notation to abbreviate summations, provided there's a one unit increment pattern in the summands.

E.g.:

$$1^3 + 2^3 + 3^3 + 4^3 + 5^3 + 6^3 + 7^3 = \sum_{n=1}^7 n^3.$$

E.g.:

$$\sqrt{3} + \sqrt{4} + \sqrt{5} + \cdots + \sqrt{88} = \sum_{k=3}^{88} \sqrt{k} = \sum_{i=1}^{86} \sqrt{i+2}.$$

This example shows that the expression of the sigma notation for a sum is not unique.

PROPERTIES OF SUMS

The following properties of sigma notation are immediate consequences of properties of the real numbers. They are very useful for manipulating and simplifying expressions involving sigma notations.

PROPERTIES OF SIGMA NOTATIONS

Let $(a_1, a_2, a_3, \dots)$ and $(b_1, b_2, b_3, \dots)$ be sequences. Then for any positive integer n and real number c , we have

- $\sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k.$
- $\sum_{k=1}^n (a_k - b_k) = \sum_{k=1}^n a_k - \sum_{k=1}^n b_k.$
- $\sum_{k=1}^n (ca_k) = c \left(\sum_{k=1}^n a_k \right).$

ACTIVITY 3

(a) Evaluate the sums below.

(i) $\sum_{k=0}^5 \frac{1}{k^2 + 1}$

(ii) $\sum_{k=1}^3 k2^k$

(b) Write the sums below using sigma notation.

(i)

$$2 + 5 + 8 + \cdots + 29.$$

(ii)

$$\frac{1}{2 \ln 3} - \frac{1}{3 \ln 4} + \frac{1}{4 \ln 5} - \frac{1}{5 \ln 6} + \cdots + \frac{1}{100 \ln 101}.$$

(iii)

$$1 - 2x + 3x^2 - 4x^3 + 5x^4 + \cdots - 100x^{99}.$$

ACTIVITY 3: SOLUTION

(a) (i)

$$\begin{aligned}\sum_{k=0}^5 \frac{1}{k^2 + 1} &= \frac{1}{0^2 + 1} + \frac{1}{1^2 + 1} + \frac{1}{2^2 + 1} + \frac{1}{3^2 + 1} + \frac{1}{4^2 + 1} + \frac{1}{5^2 + 1} \\ &= 1 + \frac{1}{2} + \frac{1}{5} + \frac{1}{10} + \frac{1}{17} + \frac{1}{26} \\ &= \frac{4193}{2210}.\end{aligned}$$

(ii)

$$\begin{aligned}\sum_{k=1}^3 k2^k &= 1 \cdot 2^1 + 2 \cdot 2^2 + 3 \cdot 2^3 \\ &= 2 + 8 + 24 = 34.\end{aligned}$$

ACTIVITY 3: SOLUTION

- (b) (i) We see that the summands have a constant increment of 3, so

$$2 + 5 + 8 + \cdots + 29 = \sum_{k=0}^9 (2 + 3k).$$

- (ii) We'll take care of the alternating sign with the factor $(-1)^k$, so

$$\begin{aligned} & \frac{1}{2 \ln 3} - \frac{1}{3 \ln 4} + \frac{1}{4 \ln 5} - \frac{1}{5 \ln 6} + \cdots + \frac{1}{100 \ln 101} \\ &= \sum_{k=2}^{100} \frac{(-1)^k}{k \ln(k+1)}. \end{aligned}$$

(iii)

$$\begin{aligned} & 1 - 2x + 3x^2 - 4x^3 + 5x^4 + \cdots - 100x^{99} \\ &= \sum_{k=0}^{99} (-1)^k (k+1)x^k. \end{aligned}$$

OUTLINE

① Sequences and Summation Notations

② Arithmetic Sequences

③ Geometric Sequences and series

④ The Binomial Theorem

⑤ Summary

PARTIAL SUMS OF ARITHMETIC SEQUENCES

When the famous mathematician **C. F. Gauss** was a schoolboy, his teacher posed the problem

$$1 + 2 + 3 + 4 + \cdots + 100$$

to the class and expected that it would keep the students busy for a long time. But Gauss answered the question almost immediately. His idea was this:

$$\begin{array}{r} S = 1 + 2 + 3 + \cdots + 98 + 99 + 100 \\ S = 100 + 99 + 98 + \cdots + 3 + 2 + 1 \\ \hline 2S = 101 + 101 + 101 + \cdots + 101 + 101 + 101 \end{array}$$

Since there are 100 terms, so there are 100 copies of 101. Hence, $2S = 100(101) \Rightarrow S = 5050$. This idea can be generalized to derive the formula of the partial sums of an arithmetic sequence.

PARTIAL SUMS OF ARITHMETIC SEQUENCES

Let S_n be the n th partial sum of the arithmetic sequence $(a, a + d, a + 2d, \dots)$, i.e. S_n is the sum of the first n terms of the arithmetic sequence:

$$S_n = a + (a + d) + (a + 2d) + \cdots + (a + (n - 1)d).$$

Using Gauss's idea, we have

$$\begin{array}{r} S_n = \quad a \quad + \quad (a + d) \quad + \cdots + [a + (n - 2)d] + [a + (n - 1)d] \\ S_n = [a + (n - 1)d] + [a + (n - 2)d] + \cdots + (a + d) + a \\ \hline 2S_n = [2a + (n - 1)d] + [2a + (n - 1)d] + \cdots + [2a + (n - 1)d] + [2a + (n - 1)d] \end{array}$$

Since there are n identical terms $2a + (n - 1)d$, so

$$2S_n = n(2a + (n - 1)d) \Rightarrow S_n = \frac{n}{2}(2a + (n - 1)d).$$

Since $a_1 = a$ and $a_n = a + (n - 1)d$, so we can simplify S_n as

$$S_n = \frac{n}{2}(a_1 + a_n).$$

PARTIAL SUMS OF ARITHMETIC SEQUENCES

PARTIAL SUMS OF AN ARITHMETIC SEQUENCE

For the arithmetic sequence with first term a and common difference d , then the n th partial sum

$$S_n = a + (a + d) + (a + 2d) + \cdots + (a + (n - 1)d)$$

is given by

$$S_n = \frac{n}{2} (2a + (n - 1)d) = \frac{n}{2} (a_1 + a_n).$$

E.g.: To find $3 + 7 + 11 + 15 + \cdots + 159$, which is a partial sum of the arithmetic sequence with $a = 3$ and $d = 7 - 3 = 4$, we first find the number of terms in the sum, n . We have
 $159 = 3 + (n - 1)(4) \Rightarrow n = 40$. So

$$3 + 7 + 11 + 15 + \cdots + 159 = S_{40} = \frac{40}{2} (3 + 159) = 3240.$$

ACTIVITY 4

- (a) A drive-in theater has spaces for 20 cars in the first parking row, 22 in the second, 24 in the third, and so on. If there are 21 rows in the theater, find the number of cars that can be parked.

- (b) Evaluate $\sum_{k=3}^{2024} (5k - 2)$.

ACTIVITY 4: SOLUTION

- (a) Let a_n be the number of car spaces in the n th row, then $(a_n)_{n=1}^{\infty}$ is an arithmetic sequence with $a = 20$ and $d = 2$. Since there are 21 rows, so the total number of cars that can be parked is

$$S_{21} = \frac{21}{2}(2(20) + (21 - 1)(2)) = 840.$$

- (b) Since the summands increase by 5 each time we increase the summation index k by 1, they have a common difference 5, with the first term $a = 5(3) - 2 = 13$. Also, there are $2024 - 3 + 1 = 2022$ terms, so

$$\begin{aligned}\sum_{k=3}^{2024} (5k - 2) &= S_{2022} = \frac{2022}{2}(2(13) + (2022 - 1)(5)) \\ &= 10242441.\end{aligned}$$

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GEOMETRIC SEQUENCES/PROGRESSIONS

Instead of adding, a geometric sequence/progression is a sequence obtained by multiplying a fixed non-zero constant r to the first term a , over and over again.

DEFINITION:

A **geometric sequence** is a sequence of the form

$$(a, ar, ar^2, ar^3, \dots).$$

The number a is the **first term**, and r is the **common ratio** of the sequence. The **n th term** of the geometric sequence above is given by

$$a_n = ar^{n-1}.$$

E.g.: If $a = 2$ and $r = 3$, then the geometric sequence is $(2, 6, 18, 54, \dots)$. Its n th term is $a_n = 2(3^{n-1})$.

E.g.: Given a geometric sequence $(2, -10, 50, -250, 1250, \dots)$, then $a = 2$ and $r = \frac{-10}{2} = -5$. Hence, its n th term is $a_n = 2(-5)^{n-1}$.

PARTIAL SUMS OF GEOMETRIC SEQUENCES

To derive the formula for the partial sums of the geometric sequence $(a, ar, ar^2, ar^3, \dots)$, i.e.

$$S_n = a + ar + ar^2 + ar^3 + \dots + ar^{n-1},$$

we multiply S_n by r and subtract from S_n :

$$\begin{array}{r} S_n = a + ar + ar^2 + ar^3 + ar^4 + \dots + ar^{n-1} \\ rS_n = \quad ar + ar^2 + ar^3 + ar^4 + \dots + ar^{n-1} + ar^n \\ \hline S_n - rS_n = a - ar^n \end{array}$$

$$\begin{aligned} \text{So} \quad S_n(1-r) &= a(1-r^n) \\ S_n &= \frac{a(1-r^n)}{1-r} \quad r \neq 1 \end{aligned}$$

Qn: The condition $r \neq 1$ is very important, otherwise we are dividing by 0. So what should be the formula when $r = 1$?

ACTIVITY 5

- (a) A ball is dropped from a height of 9 ft. The elasticity of the ball is such that it always bounces up $\frac{1}{3}$ the distance it has fallen.
- (i) Find the total distance the ball has traveled at the instant it hits the ground the fifth time.
 - (ii) Find a formula for the total distance the ball has traveled at the instant it hits the ground the n th time.
- (b) Evaluate $\sum_{k=1}^5 3 \left(-\frac{2}{3}\right)^{k-1}$.

ACTIVITY 5: SOLUTION

- (a) Let a_n be the distance traveled by the ball between the n th and $(n + 1)$ th hits. For example, when the ball first hits the ground, it bounces up to $\frac{1}{3} \times 9 = 3$ ft, and then falls for a distance of 3 ft before hitting the ground the 2nd time. So $a_1 = 3 + 3 = 6$ ft. With the same reasoning, $a_2 = \frac{1}{3}a_1 = 2$ ft, and in general $a_{n+1} = \frac{1}{3}a_n$, thus $(a_n)_{n=1}^{\infty}$ is a geometric sequence with $a = 6$ and $r = \frac{1}{3}$.
- (i) The total distance the ball has traveled at the instant it hits the ground the fifth time is

$$\begin{aligned} d &= 9 + a_1 + a_2 + a_3 + a_4 \\ &= 9 + S_4 \\ &= 9 + 6 \cdot \frac{1 - \left(\frac{1}{3}\right)^4}{1 - \frac{1}{3}} \\ &= \frac{161}{9} = 17 + \frac{8}{9}. \end{aligned}$$

ACTIVITY 5: SOLUTION

- (a) (ii) The total distance the ball has traveled at the instant it hits the ground the n th time is

$$\begin{aligned}d &= 9 + a_1 + a_2 + a_3 + a_4 + \cdots + a_{n-1} \\&= 9 + S_{n-1} \\&= 9 + 6 \cdot \frac{1 - \left(\frac{1}{3}\right)^{n-1}}{1 - \frac{1}{3}} \\&= 9 + 9 \left(1 - \left(\frac{1}{3}\right)^{n-1}\right) = 9 \left(2 - \left(\frac{1}{3}\right)^{n-1}\right).\end{aligned}$$

- (b) The corresponding geometric sequence has $a = 3 \left(-\frac{2}{3}\right)^0 = 3$ and $r = -\frac{2}{3}$. Hence,

$$\sum_{k=1}^5 3 \left(-\frac{2}{3}\right)^{k-1} = S_5 = 3 \cdot \frac{1 - \left(-\frac{2}{3}\right)^5}{1 - \left(-\frac{2}{3}\right)} = \frac{55}{27}.$$

OUTLINE

- ① Sequences and Summation Notations
- ② Arithmetic Sequences
- ③ Geometric Sequences and series
- ④ The Binomial Theorem
- ⑤ Summary

EXPANDING $(a + b)^n$

An expression of the form $a + b$ is called a **binomial**. In this section we present the formula that gives the expansion of $(a + b)^n$ for any natural number n . We first recall the definition of factorial.

DEFINITION:

The product of the first n non-zero natural numbers is called **n factorial** and is denoted by $n!$, i.e.

$$n! := 1 \cdot 2 \cdot 3 \cdot 4 \cdot \dots \cdot (n - 1) \cdot n.$$

We also define $0! := 1$.

DEFINITION:

Let n and r be natural numbers with $r \leq n$. The **binomial coefficient** is denoted by $\binom{n}{r}$ and is defined by

$$\binom{n}{r} := \frac{n!}{r!(n - r)!}.$$

KEY PROPERTY OF THE BINOMIAL COEFFICIENTS

Fun fact: In **combinatorics**, the binomial coefficient $\binom{n}{r}$ is the number of ways to choose a collection of r objects from a fixed set of n objects.

PROPERTIES OF BINOMIAL COEFFICIENTS

Let n and r be natural numbers with $r \leq n$, then the binomial coefficients have the following properties:

-

$$\binom{n}{r} = \binom{n}{n-r}.$$

-

$$\binom{n}{r-1} + \binom{n}{r} = \binom{n+1}{r}, \quad \text{for } r \geq 1.$$

Please refer to page 882-883 of the textbook for the proof.

Remark: The binomial coefficients and their properties can be derived from the **Pascal's triangle**.

THE BINOMIAL THEOREM

THE BINOMIAL THEOREM

For any positive integer n , we have

$$\begin{aligned}(a + b)^n &= a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{r}a^{n-r}b^r + \cdots + \binom{n}{n-1}ab^{n-1} + b^n \\ &= \sum_{r=0}^n \binom{n}{r}a^{n-r}b^r.\end{aligned}$$

Please refer to page 885 of the textbook for the proof.

E.g.:

$$\begin{aligned}(x + y)^4 &= x^4 + \binom{4}{1}x^3y + \binom{4}{2}x^2y^2 + \binom{4}{3}xy^3 + y^4 \\ &= x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4.\end{aligned}$$

E.g.:

$$\begin{aligned}(x - y)^4 &= x^4 + \binom{4}{1}x^3(-y) + \binom{4}{2}x^2(-y)^2 + \binom{4}{3}x(-y)^3 + (-y)^4 \\ &= x^4 - 4x^3y + 6x^2y^2 - 4xy^3 + y^4.\end{aligned}$$

ACTIVITY 6

(a) Expand $(1 - x)^5$.

(b) Simplify

$$\frac{(x + h)^3 - x^3}{h}.$$

(c) (i) Expand $\left(x^2 - \frac{1}{x}\right)^{25}$ in sigma notation.

(ii) Hence, find the coefficient of x^8 in the expansion of $\left(x^2 - \frac{1}{x}\right)^{25}$.

(d) Evaluate

$$\sum_{r=0}^n \binom{n}{r}.$$

ACTIVITY 6: SOLUTION

$$\begin{aligned} \text{(a)} \quad (1-x)^5 &= 1^5 + \binom{5}{1}1^4(-x) + \binom{5}{2}1^3(-x)^2 + \binom{5}{3}1^2(-x)^3 + \binom{5}{4}1(-x)^4 + (-x)^5 \\ &= 1 - 5x + 10x^2 - 10x^3 + 5x^4 - x^5. \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \frac{(x+h)^3 - x^3}{h} &= \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} \\ &= \frac{3x^2h + 3xh^2 + h^3}{h} = 3x^2 + 3xh + h^2. \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad \text{(i)} \quad \left(x^2 - \frac{1}{x}\right)^{25} &= \sum_{r=0}^{25} \binom{25}{r} (x^2)^{25-r} \left(-\frac{1}{x}\right)^r \\ &= \sum_{r=0}^{25} \binom{25}{r} x^{50-2r} (-1)^r \cdot x^{-r} \\ &= \sum_{r=0}^{25} (-1)^r \binom{25}{r} x^{50-3r}. \end{aligned}$$

ACTIVITY 6: SOLUTION

(c) (ii) Since

$$\left(x^2 - \frac{1}{x}\right)^{25} = \sum_{r=0}^{25} (-1)^r \binom{25}{r} x^{50-3r},$$

for x^8 , we need $50 - 3r = 8 \Leftrightarrow r = \frac{50-8}{3} = 14$. So the coefficient of x^8 is

$$(-1)^{14} \binom{25}{14} = \frac{25!}{14!11!} = 4457400.$$

(d)

$$\begin{aligned}\sum_{r=0}^n \binom{n}{r} &= \sum_{r=0}^n \binom{n}{r} 1^{n-r} 1^r \\ &= (1+1)^n = 2^n.\end{aligned}$$

OUTLINE

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SUMMARY

- Sequences and Summation Notations
 - sequences, finding the terms of a sequence, the graph of a sequence.
 - recursively defined sequences.
 - the partial sum of a sequence, the sigma notation, properties of sums.
- Arithmetic Sequences
 - definition of arithmetic sequences.
 - partial sums of arithmetic sequences.
- Geometric Sequences
 - definition of geometric sequences.
 - partial sums of geometric sequences.
- The Binomial Theorem
 - expanding $(a + b)^n$.
 - the binomial coefficients and their key properties.
 - The Binomial Theorem.

RESOURCES

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) sections 12.1, 12.2, 12.3, 12.6. Sections 12.4, 12.5 are not covered in this course, but it's good to know.